

Final Report: Adult Income Dataset Analysis

In this project, the Adult Income dataset was analyzed to predict whether an individual earns more than 50K annually. The dataset contains 48,842 records with a mix of numerical and categorical features such as age, education, workclass, occupation, and income.

The analysis began with data understanding using functions like `info()` and `describe()`, which helped identify data types and distributions. Although no missing values were initially detected, it was observed that some categorical columns contained “?” values, which were treated as missing data and removed to ensure data quality. Duplicate records were also identified and removed to avoid redundancy.

Outlier detection was performed using the IQR method. Instead of blindly removing outliers, a more appropriate approach was taken. For highly skewed features such as capital-gain and capital-loss, logarithmic transformation was applied to reduce skewness. For features like age, mild outliers were handled using capping (clipping) to preserve important data while limiting extreme values. The `fnlwgt` column was dropped as it does not contribute meaningful information for prediction.

Categorical features were converted into numerical form using one-hot encoding, while the target variable (income) was encoded into binary values (0 and 1). Numerical features such as age, educational-num, capital-gain, capital-loss, and hours-per-week were normalized using `StandardScaler` to ensure consistent scaling across features.

Data visualization was performed using univariate, bivariate, and multivariate analysis. Histograms and count plots were used to understand feature distributions. Boxplots were used to analyze relationships between variables such as age and income, showing that individuals with higher income tend to be older on average. Additional visualizations, including scatter plots and pair plots, were used to explore relationships between numerical features. A correlation heatmap was generated using selected features to avoid clutter, revealing that capital-gain and educational level have a stronger influence on income compared to other features.

From the analysis, it is evident that education level, marital status, relationship status, and capital gain play significant roles in determining income levels. Individuals who are married or have higher educational attainment are more likely to earn above 50K. The

dataset also shows class imbalance, with more individuals earning less than or equal to 50K.

Overall, the project demonstrates a complete data preprocessing and analysis pipeline, including data cleaning, transformation, outlier handling, feature engineering, and visualization. These steps what we did from 1st experiement till 9th

Start coding or generate with AI.